RANIA HAAS

CINCINNATI, OH • (419) 261-3160 • haasr3@mail.uc.edu
LinkedIn | GitHub | Portfolio

EDUCATION

University of Cincinnati **May 2027**
B.S. Computer Science | Certification in Japanese Language and Culture **GPA: 3.0**

SKILLS & CERTIFICATIONS

Programming: C++, C/AL , Svelte, HTML, JavaScript, Python, SQL, CSS, LabVIEW
Software & IDE: DynamicsNAV, Visual Studio Code, Fritizing, PlatformIO, KiCAD, Fusion 360, Ubuntu
Certifications: JLPT N4, National Association of Rocketry (NAR) L1

PROJECTS

Senior Capstone – IREC | Secretary & Payload Electronics Development Lead **August 2026 – Present**

- Served as secretary, managing and organizing the group while simultaneously leading as payload electronics development lead for Bearcat Rocketry's 10,000 ft class entry for the International Rocketry Engineering Competition(IREC) to compete in June, 2027.
- Implementing a LoRa RF telemetry link to live-stream real-time video, structural and flight data to a custom ground station build around an ESP32S3.
- Lead electronic development of a smart-rocket structural-health monitoring payload using an STM32F3203 for data acquisition, and well as sampling distributed strain gauges throughout fins and threaded rods, accelerometer, barometer, altimeter, and temperature sensors/

30,000 ft Two-Stage HPR – Custom Avionics Lead **May 2026 - Present**

- Lead and contributed on the custom avionics team, developing flight computers for two-stage deployment systems including pyrotechnic sequencing, sensor integration, redundant systems, and reliability testing.
- Built innovative multithreaded, embedded system projects and created a structured learning pipeline that introduced new team members to C and C++ fundaments, avionics concepts, and firmware development.

TOBI – Telemetry, Orientation and Barometric Instrument Flight Computer **May 2026 - Present**

- Designed and developed a custom flight computer using an Seeed ESP32-S3 and Adafruit LSM6DSO32 to monitor live data, prepare system for flight, log and record events, and enable the user to download and access system data through a front-end HTML server using C++ and with FreeRTOS.
- Refined the design using a custom PCB created in KiCAD as with as a custom 3D printed electronics bay for flight.

Harty Darty and Double Trouble – Custom Avionics Development **August 2025 – May 2026**

- Integrated code with microcontrollers, barometers, and pyrotechnics to create a custom avionics bay for Level 2 drag-separated and two stage rockets with dual redundancy via multi-threading & C++.
- Developed embedded software for launch, burnout, apogee, and descent detection, including robust data acquisition, multi-rate filtering, and structured CSV telemetry output for analysis.

NAR Level 2 Certification Rocket **Sept 2025 - Present**

- Manufactured a 4" diameter 68" tall model rocket intended for a Level 2 certification using OpenRocket to determine safety margins, develop custom fins, and custom 3D printed parts for dual deployment to achieve an apogee of 5,000ft.
- Demonstrated advanced knowledge of propulsion, avionics integration, and recovery systems in the mandated L2 test.

EXPERIENCE

McDonald's Corporate **Mar 2025 - Mar 2026**
Mentee | Chicago, Illinois (Remote)

- Worked alongside an assigned software developer mentor to develop career-oriented projects and solve industry-related problems in the Global Women in Technology Program, engaging with female leaders in corporate technology roles.

GMi Companies **Aug 2024 - Aug 2025**
Information Services Intern | Lebanon, Ohio

- Developed several code units, reports, and tables to synchronize with existing code to improve cross-referencing across the sales, processing, and other departments across on the factory floor.
- Assisted in the maintenance and optimization of Microsoft Dynamics NAV, with a focus on documentation, performance, and usability for end-users and external recipients.

ACTIVITIES

University of Cincinnati Rocketry Club **July 2025 – Present**
Member | Level 2 Mentorship Program Mentee

- Mentored by the club President for an L2 certification, building and launching a HPR rocket while developing structures, recovery systems, and safety assembly skills
- Participated in launch day standard operating procedures (SOP), flight documentation, and post flight analysis.

Seeking Full-time Positions May 2027